PulseIQ – Phase 1 Final Report

Data pipeline, quality checks, business measures, and workforce dashboard

1. Project overview

PulseIQ brings information from different files into one place. The project checks the source data, loads accepted records into a database, builds business measures, and shows workforce results in a dashboard. This report records both the work completed and the findings in the supplied data.

2. Work completed

Area	Result
Data loading	Processed 29 CSV files and 62,770 rows. Accepted 62,732 rows and set aside 38 rows for review. Row counts match.
Repeat runs	58 clean and rejected CSV outputs were identical across two runs.
Warehouse	25 tables reconciled. Evidence records 39 foreign-key constraints and no orphan rows in five sample joins.
Data quality	14 data-quality checks were implemented. All 14 SQL-backed mutation tests were executed against a separate test database, and each check detected its deliberately introduced data-quality issue. The mutation tests did not alter the production baseline.
Business measures	28 metric views were defined and exported. Six independent hand-checks matched the saved metric results.
Automated tests	The project test suite passed: 13 tests passed.
Dashboard	Four KPI cards, five charts, department and date filters, and a clear message when no data matches the filters.

3. Main validation result

The tests and saved project evidence support the data pipeline, warehouse, data-quality checks, and metric layer. A data-quality check can correctly report a problem in the source data; that does not mean the check itself failed. Source findings and limitations are described on the next page.

Data Findings and Investigation

What the source files show, without changing the data to hide a problem

4. June 2026 payroll and ledger check

The difference is INR 18,450.00 for department D01. It was traced to ledger line JL-005614 (journal JE-002094, account 5001 — Salaries & Wages). The payroll details explain the expected amounts.

Item	Amount	Check
Payroll fixed pay	INR 19,873,074.18	Expected amount for salaries and wages
Payroll variable pay	INR 10,000.00	Matches ledger account 5003
Employer PF	INR 260,158.00	Matches ledger account 5004
Payroll employer cost	INR 20,143,232.18	Payroll total
Ledger salary-related total	INR 20,161,682.18	Accounts 5001–5005
Difference	INR 18,450.00	Ledger total is higher

The bonus and PF amounts match payroll. The salaries-and-wages line is INR 18,450.00 higher than payroll fixed pay. The journal balances overall, but that does not prove the salary posting is correct. No source row or ledger entry was changed because the files do not include an approved correction.

5. Other data findings

Anomaly codes: 11 different reason codes appear in the data. The contract refers to a published list, but that official list was not included in the project files. The observed values were not treated as an approved list.

Old AI results: Five attrition predictions are dated 31 March 2026. All 405 anomaly scores have a timestamp of 1 October 2026 at 02:00 UTC. The source is a static training snapshot; timestamps were not changed and fresh predictions were not invented.

M28 forecast error: Forecasts start in October 2026, but actual headcount data ends in September 2026. There are no overlapping months, so M28 correctly reports that it cannot yet be measured.

Dashboard sources: Headcount, hires, exits, and attrition use governed metric exports. Location tier and work mode use current employee and location dimensions, as described in the dashboard task. Groups smaller than five are hidden in these two charts.

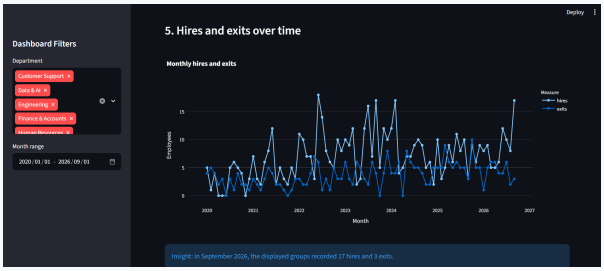
6. Final note

All checks and corrections that can be completed safely using the supplied project files are recorded. The payroll explanation and official anomaly-code list cannot be made up by changing code. The report keeps these facts visible and does not claim that source-owner approval was received.

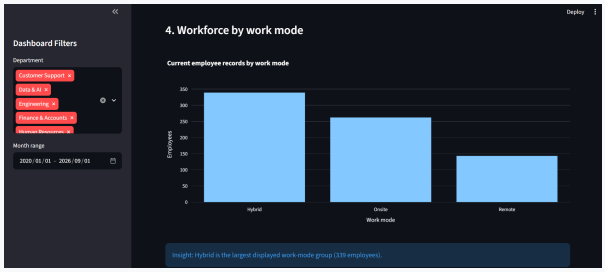
Dashboard Screenshots

Screenshots of the dashboard and its five charts

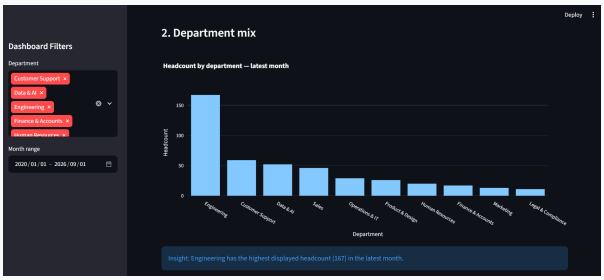
Dashboard overview



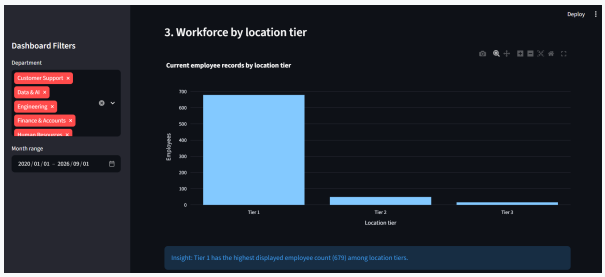
Headcount trend



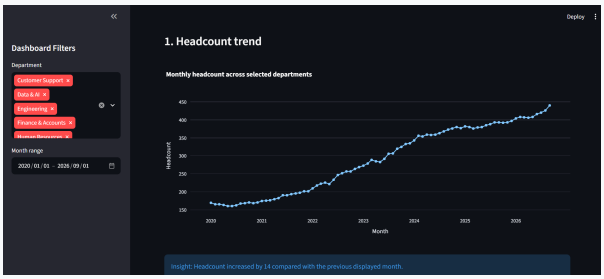
Department mix



Workforce by location tier



Workforce by work mode



Hires and exits

